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SURVEY

Text Regression Analysis: A Review, Empirical, and Experimental Insights

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ABSTRACT Effective management and analysis of large-scale textual data presents significant challenges, notably due to high storage and processing demands. Text regression analysis, a specific branch of text mining, has proven invaluable in enabling individuals, researchers, and businesses to derive meaningful insights from the rapidly increasing volumes of textual data. However, the general grouping of algorithms in existing surveys often leads to confusion and imprecise evaluations. This paper addresses these issues by introducing a methodological taxonomy tailored for text regression analysis. This taxonomy categorizes algorithms into specific techniques and detailed categories, organized into two levels: methodology category and methodology technique. To validate the accuracy of the different techniques and categories, we conduct both empirical and experimental evaluations of text regression techniques. Empirical evaluations are based on four criteria, while experimental evaluations include rankings of: (1) algorithms employing identical techniques, (2) various techniques within the same category, and (3) contrasting categories overall. This dual approach of methodological structuring and rigorous evaluation offers a nuanced and comprehensive understanding of text regression algorithms, equipping researchers with the knowledge to make informed decisions.

INDEX TERMS Text data mining, text regression analysis, experimental evaluation, empirical evaluation.

I. INTRODUCTION

Text regression analysis, a specialized area within the broader fields of text mining and natural language processing (NLP), plays a crucial role in predicting continuous numerical outcomes from textual data. This interdisciplinary domain intersects with statistics, computer science, and linguistics, addressing a variety of real-world challenges, such as quantified sentiment analysis and the estimation of document relevance over time. As our digital environment continues to expand, the ability to derive meaningful numerical predictions from unstructured text data becomes increasingly important, highlighting the significance of text regression [1].

Despite the growing relevance of this field, text regression analysis remains relatively underexplored compared to its counterpart, categorical text classification. This survey aims to bridge this gap by providing a comprehensive review of the methodologies, empirical studies, and experimental insights that define current practices in text regression and suggest directions for future research. We begin by tracing the evolution of text regression analysis, noting how technological advancements and methodological shifts have shaped its development.

The explosion of data through automated data acquisition tools, cloud storage, and Web 2.0 platforms such as Wikipedia and Twitter has dramatically increased the volume of available data, necessitating the development of effective text mining techniques [2], [3]. Text mining, which combines natural language processing, machine learning, and data mining, enables the extraction of valuable insights from this vast amount of data, with applications ranging from sentiment analysis to information retrieval [4], [5].

In this paper, we explore various algorithmic approaches, spanning from traditional statistical models to advanced machine learning techniques, all of which are tailored to predict continuous variables from textual data with accuracy. We compile and summarize empirical studies, focusing on the datasets, performance metrics, and outcomes that have

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driven the field forward. Furthermore, we synthesize experimental insights from recent applications, offering an analysis of effective strategies and common pitfalls. This synthesis is intended to guide both researchers and practitioners in their endeavors.

A. MOTIVATION AND KEY CONTRIBUTIONS

Survey papers on text regression analysis often categorize algorithms into broad groups without adequately distinguishing between the nuances of specific techniques. This generic grouping can lead to confusion, mistakenly aligning disparate algorithms and potentially resulting in misleading evaluations based on uniform metrics.

To address these challenges, we propose a detailed methodological taxonomy specifically tailored for text regression analysis. Our taxonomy hierarchically organizes algorithms into two distinct levels:

METHODOLOGY CATEGORY AND METHODOLOGY TECHNIQUE

This is the first survey paper to implement such a granular, structured taxonomy for classifying algorithms within the realm of text regression. The goal of our taxonomy is to clarify the distinctions and relationships between various algorithms by focusing on their use of specific techniques and overarching categories. This structured approach enables researchers to better understand how different algorithms relate to one another and to identify the specific techniques that are most suitable for types of text regression tasks.

In addition to presenting a detailed taxonomy, this study includes both **empirical and experimental** evaluations of text regression algorithms. We systematically compare:

1) Algorithms that utilize the same specific technique,
2) Different techniques within the same category, and
3) Different overarching methodology categories.

This layered comparison allows us to highlight the subtle yet critical distinctions between closely related algorithms and techniques. By elucidating these differences, our taxonomy and evaluations make it easier for researchers to choose the appropriate text regression approach for their specific needs.

Utilizing our methodological taxonomy, coupled with empirical and experimental insights, researchers can gain a more nuanced and comprehensive understanding of text regression algorithms, thus facilitating more informed decision-making in their research endeavors.

B. OUR METHODOLOGY-BASED TAXONOMY

Figure 1 illustrates our methodology-based taxonomy that arranges text regression algorithms into detailed classes using a hierarchical structure, consisting of methodology category and methodology technique.

C. OVERVIEW AND FORWARD-LOOKING PERSPECTIVE

In summary, this introduction lays a robust foundation for the in-depth discussions that follow, highlighting the evolving landscape and vast potential within the realm of text regression analysis. This survey not only provides an essential resource for those new to the field but also serves as a strategic roadmap for seasoned researchers, encouraging a deeper investigation into the complexities and opportunities that this important area of study presents. We aim to illuminate the path forward, inspiring continued innovation and discovery in the pursuit of more advanced and effective text regression techniques.

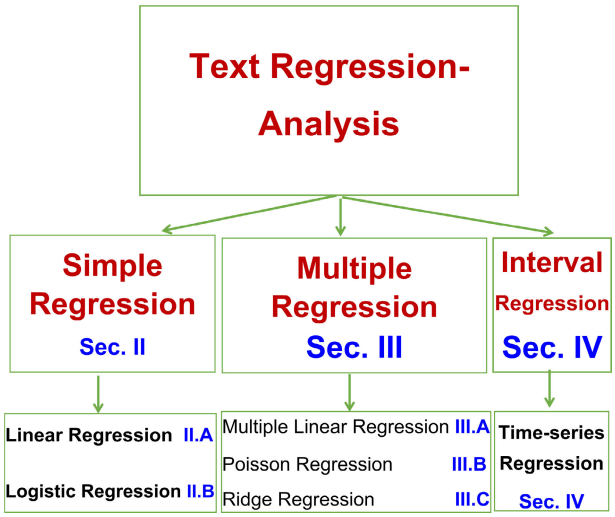

FIGURE 1. Our methodology-based taxonomy, which arranges text regression algorithms into detailed classes using a hierarchical structure, consisting of methodology category and methodology technique.

II. SIMPLE TEXT REGRESSION

A. LINEAR TEXT REGRESSION

This technique models the relationship between a dependent variable (y) and one or more independent variables (x), where y can represent various outcomes such as a text variable (e.g., sentiment score). Independent variables may encompass a wide range of features, including document length, word or sentence count, the presence of specific keywords, word frequencies, character n-grams, or syntactic patterns. These elements are crucial in determining the frequency of a particular word or term in a document or corpus. Figure 2 demonstrates this process.

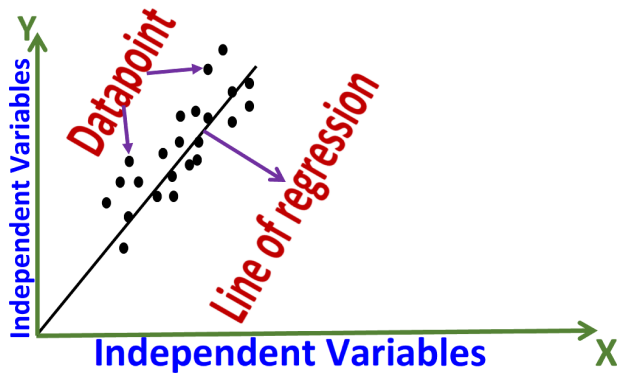

FIGURE 2. The figure depicts the process of linear regression.

The linear regression model can be represented as shown in Equation 1:

$$y = \beta_0 + \beta_1 x + \epsilon \tag{1}$$

- x represents just one predictor.
- The coefficient β_1 represents the change in y for a one-unit change in x, assuming all other factors are constant
- ϵ is the error term, accounting for the variability in y that cannot be explained by the x variables.

Linear regression is based on several key assumptions: (1) Linearity: The relationship between X and Y is linear, (2) Independence: Observations are independent of each other, (3) Homoscedasticity: The variance of residual is the same for any value of X, and (4) Normal Distribution of Residuals: For any fixed value of X, Y is normally distributed.

1) THE RATIONALE BEHIND THE USAGE OF THE TECHNIQUE

By analyzing the correlation between predictor variables and outcomes, one can achieve three objectives: (1) examine the impact of each variable on the outcome while controlling for others, (2) develop a predictive model based on the values of predictor variables, (3) identify outcome prediction.

2) CONDITIONS FOR THE OPTIMAL PERFORMANCE OF THE TECHNIQUE

To enhance the technique: (1) maintain proportional changes between dependent and independent variables; (2) avoid high degree of correlation between independent variables; (3) ensure normal distribution of difference between predicted and actual values; (4) consistent residual variance across all independent variable values.

3) LIMITATIONS OF THE TECHNIQUE

Limitations include: inappropriate use when a non-linear relationship exists between variables; inaccurate estimates when predictor variables are highly correlated; sensitivity to outliers; violation of assumption of independence between predictor variables due to high correlation in text mining; and limited applicability to categorical variables.

4) FEATURING RESEARCH ARTICLES THAT HAVE IMPLEMENTED THE TECHNIQUE

Cendrero et al. [8] proposed a topic model that combines Latent Dirichlet Allocation and linear regression through an elastic net. Georgieva-Trifonova et al. [9] proposed a framework for systematic literature review. It involves the use of linear regression for examining trends. Ye et al. [10] classified web services using linear regression algorithm to combine the predictions of web service category breadth and depth into the result.

Samara et al. [11] examined crime data to identify the primary factors influencing the likelihood of crime. The variables analyzed in this study include socio-economic factors such as race, household size, and average household income. McClendon et al. [12] utilized data obtained from neighborhoodscout.com and the University of California-Irvine, in Mississippi, to forecast crime patterns through additive linear regression.

Freo et al. [13] explored how analytics can leverage the informative content and explanatory potential of unstructured, short texts on a response variable, using linear regression models to assess the marginal contribution of attributes. They justified their focus on regression analysis for ratings for two reasons: firstly, variable selection methods for categorical data are less advanced than those for continuous variables, complicating direct comparisons; secondly, ratings are commonly treated as quantitative variables, enhancing the applicability of their findings across various related domain.

TABLE 1. Evaluating research papers that have employed linear regression-based techniques.

Paper	Dataset	Scalability	Interpretability	Accuracy	Efficiency
[8] 2022	Author Collected	Unsatisfactory	Good	Acceptable	Fair
[9] 2022	Author Collected	Fair	Acceptable	Fair	Fair
[10] 2019	Author Collected	Fair	Good	Acceptable	Fair
[11] 2023	Communities and Crime Unnormalized dataset, UCI	Unsatisfactory	Good	Good	Fair
[12] 2015	Communities and Crime Unnormalized dataset, UCI	Unsatisfactory	Fair	Fair	Fair
[13] 2024	Knitwear and Dresses dataset	Fair	Acceptable	Acceptable	Fair

B. LOGISTIC TEXT REGRESSION

This technique employs a logistic function, also known as a sigmoid function, to model the relationship between independent variables (e.g., word frequencies) and a dependent variable, effectively mapping inputs to probabilities. It transforms the input features into a probability output, which is limited to the range of 0 to 1 by the logistic function, making it suitable for resolving binary classification tasks. Coefficients within this model indicate the change in log odds of the dependent variable for a unit change in the independent variable, encapsulating the essence of how each predictor influences the probability outcome. Figure 3 demonstrates this procedure.

The logistic function is defined as shown in Equation 2:

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-z}} \tag{2}$$

where z is a linear combination of the input features (X), $z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n$, and β are the parameters

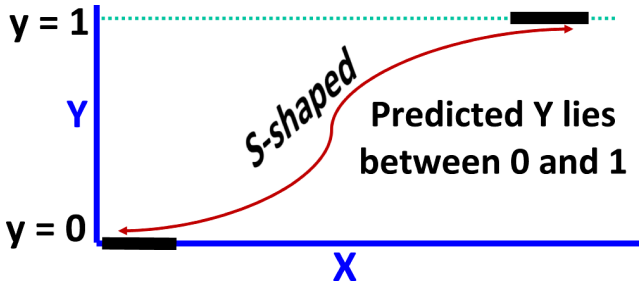

FIGURE 3. The figure depicts the process of logistic regression.

of the model. The parameters (β) of the logistic regression model are typically estimated using maximum likelihood estimation. This method tries to find the parameter values that maximize the likelihood of the observed sample.

1) THE RATIONALE BEHIND THE USAGE OF THE TECHNIQUE

Associating input features with a probability output brings about several benefits, including (1) a simple and easy-to-understand algorithm, (2) effectiveness with data that has high dimensionality, (3) the ability to handle non-linear connections, (4) output that is simple to interpret, and (5) preventing overfitting and enhance generalization.

2) CONDITIONS FOR THE OPTIMAL PERFORMANCE OF THE TECHNIQUE

It can be enhanced by ensuring balanced classes, as an imbalanced distribution can lead to bias towards the majority class and affect the model's performance on the minority class. The training data should be independent and identically distributed. Only relevant features for the classification should be selected. Avoiding overfitting by regularization.

3) LIMITATIONS OF THE TECHNIQUE

(1) linearity assumption, which may not hold for complex, (2) feature selection challenge due to many features, leading to overfitting, (3) difficulty in accurately classifying the minority class in imbalanced data, resulting in lower accuracy, and (4) multicollinearity.

4) FEATURING RESEARCH ARTICLES THAT HAVE IMPLEMENTED THE TECHNIQUE

Ginting et al. [14] used Multinomial Logistic Regression to classify hate speech elements in text. Wen et al. [15] used film reviews, where text features were extracted using TFID+LSA, and logical regression was used to analyze the features. Al-Aamri et al. [16] presented a text mining method for constructing a gene-gene interaction network using weighted logistic regression to identify related genes.

Schwarz et al. [17] introduced commands for estimating text regressions applicable to continuous, binary, and categorical variables. The command txtreg_train streamlines text cleaning, tokenization, model training, and cross-validation for methods like lasso, ridge, elastic-net, and regularized logistic regressions. txtreg_predict is used for generating model predictions, while txtreg_analyze helps in analyzing model coefficients. These tools collectively simplify the process of training text regression models and facilitate the sharing of pretrained models among researchers.

Wen et al. [15] introduced a method for analyzing text emotion using a logistic regression model. They utilized 99,452 lines of film review data from web pages to extract text features using TFIDF+LSA. The features were analyzed using logistic regression and Bayesian models, and accuracy was validated through AUC of the reserve method. The findings demonstrated that the classical logistic regression model is cost-effective for emotion classification.

Li et al. [18] introduced the Logistic Regression Matching Pursuit (LRMP) algorithm for text classification, featuring a novel loss function and residual update function. This algorithm efficiently minimizes the negative log likelihood in just one iteration, with its performance underpinned by the strong Wolfe condition. This ensures optimized classification accuracy while significantly accelerating the training process.

TABLE 2. Evaluating research papers that have employed logistic regression-based techniques.

Paper	Dataset	Scalability	Interpretability	Accuracy	Efficiency
[14] 2019	Author Collected	Fair	Unsatisfactory	Acceptable	Acceptable
[15] 2022	Author Collected	Acceptable	Unsatisfactory	Acceptable	Fair
[16] 2017	STRING dataset	Fair	Fair	Acceptable	Acceptable
[17] 2023	Author Collected	Unsatisfactory	Unsatisfactory	Fair	Fair
[18] 2022	Author Collected	Unsatisfactory	Unsatisfactory	Acceptable	Fair
[18] 2023	Amazon product reviews and movie reviews datasets	Fair	Unsatisfactory	Good	Good

III. MULTIPLE LINEAR TEXT REGRESSION

A. MULTIPLE LINEAR TEXT REGRESSION

The technique predicts the occurrence of a word or phrase in a document and numerical outcomes from text based on the presence or absence of other words or phrases. The process involves collecting data and transforming text data into numerical data through feature extraction methods, such as Bag of Words or TF-IDF. After feature extraction, the data is split into training and testing sets.

A statistical software like R is then used to train a multiple linear regression model on the training data. The model's performance is assessed, and optimization strategies, including

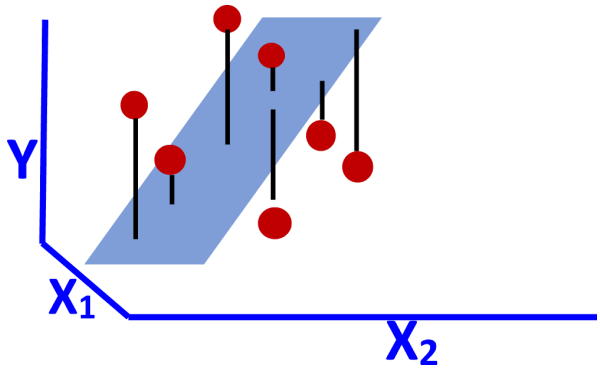

FIGURE 4. The figure depicts the process of multiple linear regression.

adjusting hyperparameters, regularization, or modifying feature extraction methods, are applied to enhance the model's accuracy.

The general form of the multiple linear regression equation is as shown in Equation 3:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n + \epsilon \tag{3}$$

- y is the dependent variable.
- $x_1, x_2, \ldots, x_n$ represent multiple predictors.
- β_0 is the intercept. It represents the value of y when all independent variables are zero.
- $\beta_1, \beta_2, \ldots, \beta_n$ are the coefficients of the independent variables $x_1, x_2, \ldots, x_n$.
- Each coefficient β_i represents the change in y for a one-unit change in x_i, assuming all other predictors are held constant.
- ϵ is the error term, accounting for the variability in y not explained by the independent variables.

1) THE RATIONALE BEHIND THE USAGE OF THE TECHNIQUE

Using linear regression modeling, a group of predictor variables (i.e., words or phrases) can be analyzed in relation to a dependent variable (e.g., topic or sentiment) to determine the strength of their association. Determining which predictor variables are influential in a topic or sentiment provides an understanding of the crucial words or phrases.

2) CONDITIONS FOR THE OPTIMAL PERFORMANCE OF THE TECHNIQUE

The technique can be improved by: (1) Adjusting hyperparameters (e.g., fine-tuning regularization parameters), (2) Adhering to linear regression assumptions, (3) Validating the model (using metrices such as MAE and MSE), (4) Avoiding highly correlated features), and (5) Scaling features (normalizing features for uniform scaling).

3) LIMITATIONS OF THE TECHNIQUE

Limitations include the linearity assumption that can result in suboptimal results if the relationship is nonlinear. Techniques like TF-IDF may not capture semantic relationships. High-dimensionality in text can lead to overfitting. Multicollinearity can reduce model interpretability when highly correlated features are present. Sparse data can pose challenges in training.

4) FEATURING RESEARCH ARTICLES THAT HAVE IMPLEMENTED THE TECHNIQUE

Munkhdalai et al. [19] proposed a locally adaptive interpretable regression method that enhances the linear multivariate regression model with recurrent neural networks. Chen et al. [20] combined multivariate adaptive regression splines and queen genetic algorithm to predict all-inclusive financial metrics related to operating revenue. Wei et al. [21] proposed a percentile variable selection method for multivariate regression to decrease the dimensions of a model, while retaining its generalization performance.

Sheng et al. [22] examined the shortcomings of current segmentation-based scene text detectors and introduced a new text instance multiple linear expression that incorporates both foreground and background information, utilizing pixels near text boundaries as offset starts. They developed a post-processing algorithm that sequentially integrates four prediction results to accurately reconstruct the text instance.

TABLE 3. Featuring evaluating research papers that have employed multiple linear regression technique.

Paper	Dataset	Scalability	Interpretability	Accuracy	Efficiency
[20] 2022	Financial dataset	Unsatisfactory	Good	Good	Fair
[21] 2019	Author Collected	Unsatisfactory	Acceptable	Good	Good
[22] 2022	Author Collected	Fair	Acceptable	Acceptable	Fair
[23] 2021	SynthText and Total-Text datasets	Unsatisfactory	Acceptable	Good	Acceptable
[24] 2015	Malaria Dataset	Fair	Good	Acceptable	Acceptable

B. POISSON TEXT REGRESSION

Poisson regression models are statistical tools used to analyze count data, particularly the probability of a specific number of events occurring within a fixed interval of time or space. This technique is applied to understand the frequency of word occurrence in documents or corpora. Poisson regression models word frequency by considering factors like document length/related terms, assuming occurrences follow a Poisson distribution, revealing usage patterns.

The basic form of the Poisson regression equation can be expressed as shown in Equation 4:

$$\log(\lambda_i) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n \tag{4}$$

where λi is the expected number of occurrences, $\beta 0, \beta 1, \ldots, \beta n$ are coefficients, and X1, X2, …, Xn are the predictor

variables. The assumptions are: (1) the mean and variance of the dependent variable are equal (equidispersion), and (2) the observations are independent of each other.

1) THE RATIONALE BEHIND THE USAGE OF THE TECHNIQUE

Using the Poisson distribution to model the frequency of events can serve two purposes: Firstly, it can help identify frequently used words, their connections, and how these evolve over time or across document types. Secondly, it can model count data that may not follow a normal distribution, which is common in textual data.

2) CONDITIONS FOR THE OPTIMAL PERFORMANCE OF THE TECHNIQUE

The technique can be enhanced by: (1) utilizing count data to show term frequency in a document, (2) ensuring term occurrences are independent, (3) maintaining consistent count variance across predictor variable levels, (4) achieving linear relationship between predictor variables and response variable log, (5) avoiding overdispersion.

3) LIMITATIONS OF THE TECHNIQUE

(1) dependence on independence assumption (words or phrases may have interdependence), (2) restriction to count data (but qualitative attributes like tone or sentiment are common), (3) overdispersion (when variance is higher than mean), (4) difficulty in interpreting coefficients (word/phrase interactions are complex and hard to measure).

4) FEATURING RESEARCH ARTICLES THAT HAVE IMPLEMENTED THE TECHNIQUE

Zhang et al. [23] suggested a Poisson regression-based technique for modeling disease epidemic diffusion patterns over time, training separate models for each year. Zhang et al. [24] proposed SDAE-PR, which uses stacked denoising autoencoders to extract high-level features of data, then applies Poisson regression to the features.

Wu et al. [25] introduced a novel sparse-constrained Poisson regression (SCPR) algorithm using a Poisson distribution model to represent word-count features in textual data. This approach employs a sparsity ratio that combines both $\ell 1$-norm and $\ell 2$-norm on the weights, facilitating feature selection and improving document clustering. To address high-dimensional data, they designed a projected gradient descent method, enhancing the SCPR algorithm's efficiency and scalability compared to traditional Poisson regression.

Aarthi et al. [26] proposed a medical image encryption method combining Poisson regression, AES-128 encryption, and Arnold CAT map to introduce randomness in RGB medical images. This technique not only reduces the time and space complexities by converting images into UTF-8 text format but also enhances security, achieving a NPCR score of 99.0174 and a UACI score of 33.0690. Future work will explore leveraging deep CNN models for medical image encryption and chaotic sequence mapping.

TABLE 4. Evaluating research papers that have employed poisson regression-based techniques.

Paper	Dataset	Scalability	Interpretability	Accuracy	Efficiency
[23] 2015	Malaria dataset	Unsatisfactory	Acceptable	Good	Fair
[24] 2022	Synthetic dataset	Unsatisfactory	Acceptable	Acceptable	Fair
[25] 2013	TDT2 document corpora	Unsatisfactory	Fair	Acceptable	Fair
[26] 2022	NIH chest x-ray images	Unsatisfactory	Unsatisfactory	Acceptable	Fair

C. RIDGE TEXT REGRESSION

This technique adds a penalty term to the traditional regression equation. This penalty term shrinks the coefficients of the predictor variables, reducing their overall impact on the response variable. The amount of shrinkage is controlled by a tuning parameter.

In ridge regression, a penalty term is added to the cost function used in ordinary least squares regression. The cost function in ridge regression is shown in Equation 5.

$$J(\Phi) = \frac{1}{2m}\sum_{i=1}^{m}\left(h_{\Phi}\left(x^{i}\right) - y^{i}\right)^{2} + \lambda\sum_{j-1}^{n}\Phi_{j}{}^{2} \tag{5}$$

where $h_{\varphi}(x)$ is the hypothesis function, φ represents the regression coefficients, m is the number of training examples, λ is the regularization parameter, and n is the number of features. The first term is the least squares cost and the second term is known as the regularization term.

The regularization term $\lambda\sum_{j-1}^{n}\Phi_{j}^{2}$ penalizes the magnitude of the coefficients, which helps to reduce overfitting. Higher values of λ increase the penalty, and as λ approaches zero, ridge regression approaches least squares regression.

The regularization parameter λ controls the impact of the penalty term on the regression coefficients. It is typically selected via cross-validation, where different values of λ are tested, and the one that results in the best model performance is chosen.

The ridge coefficients minimize the penalized residual sum of squares and can be solved using the matrix formula: $\varphi = (\mathrm{X}^{\mathrm{T}} + \lambda\mathrm{I})^{-1}\mathrm{X}^{\mathrm{T}}\,\mathrm{y}$, where I is the identity matrix of appropriate size. This adjustment to the matrix $\mathrm{X}^{\mathrm{T}}\ \mathrm{X}$ (by adding $\lambda\mathrm{I}$) ensures that it is invertible, solving issues related to multicollinearity

1) THE RATIONALE BEHIND THE USAGE OF THE TECHNIQUE

The problem of high correlation between predictor variables is commonly faced by traditional linear regression techniques, but it can be mitigated by including a penalty term in the regression equation. This causes coefficients to be reduced towards zero, decreasing the influence of highly

correlated variables on the model coefficients, (still play role in the prediction).

2) CONDITIONS FOR THE OPTIMAL PERFORMANCE OF THE TECHNIQUE

The technique can be enhanced by: (1) optimal selection of the regularization parameter (λ) through cross-validation techniques to minimize mean squared error, (2) selection of relevant features (e.g., Lasso or Elastic Net) to improve feature selection and reduce overfitting, (3) standardization of variables by scaling them to have a mean of 0 and a SD of 1.

3) LIMITATIONS OF THE TECHNIQUE

(1) High-dimensional data can pose a challenge as it expects fewer predictors than observations, which may not hold in text mining. (2) the penalty term introduces bias and leads to underfitting. (3) Ridge regression is less interpretable than other models.

4) FEATURING RESEARCH ARTICLES THAT HAVE IMPLEMENTED THE TECHNIQUE

Jin et al. [27] introduced a graph relational network for modeling high-order relationships in noisy data using ridge regression to find adjacent nodes. Mostafa et al. [28] suggested an algorithm for handling missing data using Bayesian Ridge Regression to create a predictive model.

Onita and Cucu [29] explored automatic text summarization by analyzing Facebook posts related to conspiracy theories using a Kernel Ridge Regression model. Their model outperformed a deep learning LSTM-based approach with an Attention Mechanism in generating summaries that closely mirrored the original posts, avoiding repetition and irrelevance. Experiments on artificially generated data confirmed the superior performance and qualitative effectiveness of their approach.

Liu [30] utilized Python to preprocess online comment data by refining text and organizing reviews into categories such as service and location, using the Paddlehub library to assess sentiment across these categories. Ridge Regression and k-fold cross-validation were then employed to create a comprehensive evaluation model for each scenic spot and hotel, which calculates total scores based on various metrics including. Also, a method for extracting characteristic words was developed using LDA for topic extraction and a series of text processing techniques, allowing for a nuanced comparison and classification of scenic spots and hotels.

IV. TEXT INTERVAL TIME-SERIES REGRESSION

This technique analyzes the dynamic relationship between a dependent variable and one or more independent variables, focusing on how linguistic changes over time relate to external factors like public opinion shifts or news events. It starts with text preprocessing and converts text into numerical features, often through a bag-of-words model, to study linguistic trends and their external influences over time.

TABLE 5. Evaluating research papers that have employed ridge regression-based techniques.

Paper	Dataset	Scalability	Interpretability	Accuracy	Efficiency
[27] 2023	Cora, Citeseer, Pubmed	Acceptable	Unsatisfactory	Good	Fair
[28] 2020	Author Collected	Fair	Unsatisfactory	Acceptable	Acceptab
[29] 2023	COVID-19 Open Research Dataset Challenge	Good	Unsatisfactory	Good	Acceptable
[30] 2021	Author Collected	Fair	Unsatisfactory	Acceptable	Fair

1) THE RATIONALE BEHIND THE USAGE OF THE TECHNIQUE

Time-series regression is useful for analyzing and modeling relationships between variables over time, while text mining is useful for analyzing and extracting insights from unstructured text. By combining these two approaches, we can gain a complete understanding of the data and potentially improve the accuracy of our predictions.

2) CONDITIONS FOR THE OPTIMAL PERFORMANCE OF THE TECHNIQUE

The technique can be enhanced by: (1) Obtaining adequate and pertinent data, (2) Acquiring data of high quality, (3) Ensuring stationarity by testing and transforming the data when required to meet the model's assumptions, (4) Considering autocorrelation, which is common in time-series data, (5) Choosing relevant independent variables.

3) LIMITATIONS OF THE TECHNIQUE

The limitation include the challenge of preprocessing textual data due to the complex and diverse nature of language, the difficulty of converting textual data into a meaningful representation for analysis, and the complexity that arises when dealing with high dimensional data in time-series regression models.

4) FEATURING RESEARCH ARTICLES THAT HAVE IMPLEMENTED THE TECHNIQUE

Jiang et al. [31] combined DNN and traditional time series regression models to improve the accuracy of time series forecasting. Jiang et al. [32] proposed a method for modeling consumer preferences from online feedback using nonlinear time series fuzzy regression.

Wang et al. [33] proposed a novel model for addressing the multivariate time series regression problem, commonly encountered across various fields. They introduced a general functional mapping framework, with a function-on-function linear model as a specific case, and extended it to a nonlinear model using a fully connected neural network to enhance learning directly from data. Detailed numerical

implementation procedures for this advanced model were thoroughly documented.

Azib et al. [34] introduced a novel deep-learning supervised method for event detection in multivariate time series data, featuring several distinctive characteristics compared to existing approaches. Their method, primarily focused on regression rather than binary classification, utilizes a list of ground truth events for training, incorporates a stacked ensemble learning meta-model to enhance robustness, and offers a user-friendly Python package to facilitate practical implementations. Demonstrated across diverse use cases, including NLP tasks and financial security, their approach emphasizes adaptability and ease of use, with an API that supports extensive hyperparameter optimization to potentially improve performance across various domains.

TABLE 6. Evaluating research papers that have employed time-series regression-based techniques.

Paper	Dataset	Scalability	Interpretability	Accuracy	Efficiency
[31] 2019	Author Collected	Unsatisfactory	Acceptable	Acceptable	Good
[32] 2022	Author Collected	Unsatisfactory	Fair	Good	Good
[33] 2020	Electricity demand records of Adelaide	Unsatisfactory	Acceptable	Good	Acceptable
[34] 2023	Credit card fraud dataset	Unsatisfactory	Good	Good	Good

V. TEXTEXPERIMENTAL EVALUATION

In this study, we conducted an empirical evaluation of the techniques presented within this paper. We selected a specific algorithm that utilizes each technique as its representative for comparative analysis. This approach allowed us to assess and compare the performance of these algorithms, providing an overarching view of each technique's accuracy.

It is important to note that the accuracy of each algorithm may vary depending on the chosen parameters, and thus, the results may not fully represent the accuracy. To ensure consistency, we adhered to the experimental setups and parameters detailed in the original publications of each algorithm.

The evaluations were performed on a machine equipped with Windows 10 Pro, powered by an Intel(R) Core(TM) i7-6820HQ processor with a 2.70 GHz CPU speed and 16 GB of RAM.

A. DATASETS

- **Amazon Reviews Dataset (ARD)** [35]**:** The dataset includes 34,686,770 Amazon reviews from 6,643,669 users across 2,441,053 products, sourced from the Stanford Network Analysis Project (SNAP). For our evaluation, we utilized a subset comprising 1,800,000 training samples and 200,000 testing samples for each sentiment polarity.
- **Yelp Reviews Dataset (YRD)** [36]**:** The Yelp Reviews Dataset is a large collection of over 8 million customer reviews from Yelp, a platform for finding local businesses. It covers more than 200,000 businesses and includes reviews from over 1 million unique users. Each review features a star rating (1 to 5) and a timestamp.
- **Boston Housing dataset** (**BHD**) [37] and the Energy Efficiency dataset (EE) [38]. BHD contains information collected on housing in Boston, USA. It comprises 506 observations and 13 attributes, such as crime rate, accessibility to highways, and median value of owner-occupied homes, including the average number of rooms per dwelling.
- **EE dataset**: It contains data on the heating and cooling load requirements of buildings. The dataset comprises 768 observations and 8 attributes, such as building orientation.

B. METHODOLOGY FOR SELECTING REPRESENTATIVE PAPERS

After scrutinizing papers that documented algorithms using a specific technique, we pinpointed the most influential ones. The algorithms showcased in Table 7 were selected as representatives of their respective techniques. To identify the most significant paper among those discussing algorithms that employ the same technique, we evaluated various factors, including the innovativeness of the research and the These criteria allowed us to determine the contemporary relevance and impact of each paper.

TABLE 7. List of selected papers.

Category	Technique	Algorithm
Simple Regression	Linear Regression	[8]
	Logistic Regression	[14]
Multiple Regression	Multiple Linear Regression	[19]
	Poisson Regression	[23]
	Ridge Regression	[27]
Interval Regression	Time Series Regression	[31]

We developed our own implementation using TensorFlow [39] and trained them with Adam optimizer [40], [41]. TensorFlow's APIs allow users to create their own ML algorithms [39]. We used Python 3.6 as our development language and TensorFlow 2.10.0 as the backend.

C. EVALUATION SETUP

We evaluated the performance of six regression algorithms by testing their accuracies using default parameter settings. Prior to running the analyses, data preprocessing involved replacing punctuation with spaces and omitting words that appeared only once across documents. For the optimization process, we employed the Adam optimizer, setting the batch size to 64 and the initial learning rate at 0.0005.

The efficacy of each regression method was assessed using the Root Mean Squared Error (RMSE) metric, which measures the average discrepancy between the predicted and

TABLE 8. Parameter setup.

Regression Type	Parameter	Value	Notes
Logistic Regression [14]	C	1.0	Regularization strength
	solver	'liblinear'	For binary targets
Poisson Regression [23]	max_iter	100	Convergence criterion
Ridge Regression [27]	alpha	1.0	Regularization strength
	alpha range	[0.01, 0.1, 1, 10, 100]	Possible values for alpha
Interval Time-Series Regression [31]	max_iter	100	Iterative procedure
Linear Regression [8]	L1 (Lasso) regularization	λ = 0.01	Can vary from 0.001 to 1
	L2 (Ridge) regularization	λ = 0.1	can vary from 0.001 to 1
Multiple Regression [19]	Variance Inflation Factor (VIF)	Threshold values less than 5	To decide if a feature should be dropped due to multicollinearity
	Recursive Feature Elimination (RFE)	Continue until reaching predefined 20 features	Depending on the initial number of features

actual values of a variable. Specific parameter values applied to each algorithm are detailed in Table 7.

D. RANKING THE VARIOUS TECHNIQUES, SUB-CATEGORIES, AND CATEGORIES

We ranked the techniques and categories as follows.

- ➢ We calculated the average scores of the chosen algorithms that utilized the same technique. Then, we ranked the techniques that fall under the same technique based on their scores.
- ➢ We calculated the average scores of the chosen algorithms that utilized the same category. Then, we ranked the categories based on their scores.

E. EXPERIMENTAL RESULTS

Table 9 and Figs. 5-8 depict the experimental results. Table 9 shows the RMSE values and the rankings of the different techniques and categories. Figs. 5-8 show the RMSE values using the Amazon Review Dataset, Yelp Review Dataset, Boston Housing, and EE datasets, respectively.

F. DISCUSSION OF THE EXPERIMENTAL RESULTS

Simple Regression Methods

- **Linear Regression**: The RMSE values for Linear Regression are: 8.42 for ARD, 6.84 for YRD, 3.53 for the BHD dataset. and 2.1 for the EE dataset. While Linear Regression maintains interpretability, its accuracy is somewhat limited, particularly with data features that exhibit high correlation. This issue, noted as multicollinearity, makes it difficult to interpret regression coefficients reliably.
- **Logistic Regression**: Exhibited comparable performance as Linear Regression with RMSEs of: 11.46 (ARD), 9.48 (YRD), 3.22 (BHD) and 1.67 (EE). This might suggest better handling of categorical or dichotomous outcome variables compared to the continuous outcomes in Linear Regression. However, the simple nature of Logistic Regression could still be prone to issues like underfitting in complex datasets.

Multiple Regression Methods

- **Multiple Linear Regression**: With RMSEs Of: 7.48 (ARD), 4.53 (YRD), 3.18 (BHD), and 1.62 (EE), this method shows an improvement over simple Linear Regression, likely due to its ability to handle multiple predictors. It ranks first in its category for the EE dataset, indicating robust performance.

TABLE 9. The RMSES of the regression analysis methods. the table also shows the following: (1) the ranking of the techniques that fall under a same category, and (2) the ranking of the different categories.

Category	Technique	Selected Papers	Data set	RMSE	Technique Rank	Category Rank
Simple Regression	Linear Regression	[8]	ARD	8.42	1	2
			YRD	6.84		
			BHD	3.53		
			EE	2.10		
	Logistic Regression	[14]	ARD	11.46	2	
			YRD	9.48		
			BHD	3.22		
			EE	1.67		
Multiple Regression	Multiple Linear Regression	[19]	ARD	7.48	2	1
			YRD	4.53		
			BHD	3.18		
			EE	1.62		
	Poisson Regression	[23]	ARD	9.59	3	
			YRD	6.31		
			BHD	3.32		
			EE	1.87		
	Ridge Regression	[27]	ARD	5.74	1	
			YRD	4.66		
			BHD	3.13		
			EE	1.56		
Interval Regression	Time Series Regression	[31]	ARD	10.17	N/A	3
			YRD	8.47		
			BHD	3.28		
			EE	1.76		

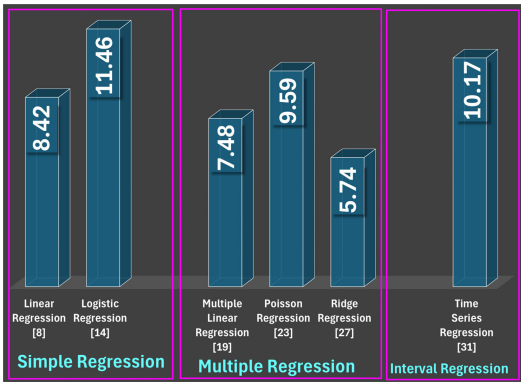

FIGURE 5. The RMSE values of the regression analysis methods using the Amazon Review Dataset (ARD).

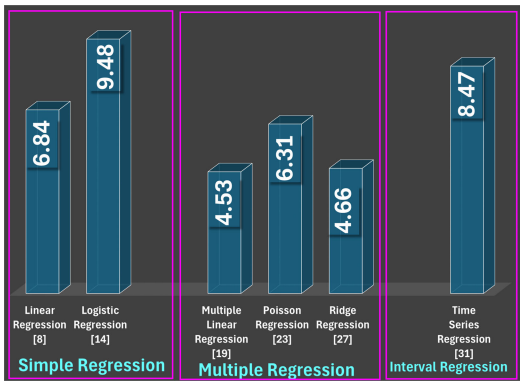

FIGURE 6. The RMSE values of the regression analysis methods using the Yelp Review Dataset (YRD).

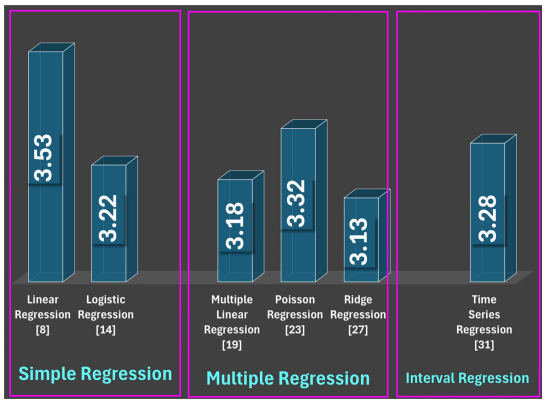

FIGURE 7. The RMSE values of the regression analysis methods using the Boston Housing dataset (BHD).

FIGURE 8. The RMSE values of the regression analysis methods using the EE dataset.

- **Poisson Regression**: Produced RMSEs of: 9.59 (ARD), 6.31 (YRD), 3.32 (BHD), and 1.87 (EE). The noted overdispersion problem—where the variance equals the mean of the predictions—suggests that Poisson Regression may not be ideal for data that do not follow a Poisson distribution closely.
- **Ridge Regression**: This technique yielded the best RMSEs among the multiple regression methods with 5.74 (ARD), 4.66 (YRD), 3.13 (BHD), and 1.56 (EE). Ridge Regression addresses multicollinearity by adding a degree of bias to the regression estimates, hence stabilizing the solutions and often leading to better long-term predictions.

Time Series Regression

- **Time Series Regression**: The RMSE for this method is 10.17 (ARD), 8.47 (YRD), 3.28 for BHD, and 1.76 for EE. The fusion of DNNs (Deep Neural Networks) with linear models has enhanced predictive precision, but its performance dips with non-stationary data due to the model's inability to dynamically adapt to changes in trend over time. The inclusion of RNN (Recurrent Neural Network) elements into the model slightly decreases RMSE but does not fully mitigate the challenges with non-stationary data.

Summary

Across the various methods:

- Ridge Regression appears most effective in handling multicollinearity and providing stable predictions.
- Poisson Regression, while useful in certain contexts, struggles with overdispersion.
- Time Series Regression benefits from hybrid models incorporating neural networks, though challenges remain with non-stationary data.
- Simple regression methods are less effective with complex, highly correlated datasets but maintain high interpretability.

VI. CHALLENGES AND POTENTIAL FUTURE PERSPECTIVES FOR TEXT REGRESSION

A. LINEAR TEXT REGRESSION

Challenges:

- ***Sparsity of Text Data:*** Text data is typically high-dimensional and sparse, which can lead to overfitting.
- ***Feature Selection:*** Choosing the right features (e.g., word counts, TF-IDF) for regression models can be non-trivial and significantly affects performance.

Future Perspectives and Improvements:

- ***Advanced Feature Engineering:*** Using more sophisticated NLP techniques like embeddings or contextual features from models like BERT.
- ***Regularization Techniques:*** Improved regularization strategies to manage high dimensionality and avoid overfitting.

B. LOGISTIC TEXT REGRESSION

Challenges:

- ***Class Imbalance:*** Common in text classification tasks, where some classes are significantly underrepresented.
- ***Non-linear Separability:*** Logistic regression assumes linear boundaries between classes, which might not hold in complex text data.

Future Perspectives and Improvements:

- ***Ensemble Methods:*** Combining multiple logistic regression models or integrating with other algorithms to handle non-linearity.

- ***Balanced Training Approaches:*** Techniques like SMOTE or adaptive resampling to address class imbalance.

C. MULTIPLE LINEAR TEXT REGRESSION

Challenges:

- ***Multicollinearity:*** High correlation between independent variables (features) can distort the estimation.
- ***Interpretability:*** With many predictors, it becomes challenging to interpret the effects of all variables.

Future Perspectives and Improvements:

- ***Dimensionality Reduction:*** Techniques like PCA or LDA to reduce features to a manageable number while retaining most of the variance.
- ***Model Explainability Tools:*** Utilizing tools and techniques to better explain model predictions and feature importances.

D. POISSON TEXT REGRESSION

Challenges:

- ***Overdispersion:*** When the variance exceeds the mean, standard Poisson regression may not be adequate.
- ***Zero Inflation:*** Excessive zeros in count data can lead to biased estimates.

Future Perspectives and Improvements:

- ***Zero-Inflated Models:*** Development of more robust zero-inflated Poisson models for better handling of excess zeros.
- ***Advanced Count Models:*** Utilizing negative binomial regression or other generalized models to address overdispersion.

E. RIDGE TEXT REGRESSION

Challenges:

- ***Bias-Variance Trade-off:*** While ridge regression reduces variance by introducing bias, finding the optimal regularization strength (lambda) is crucial.
- ***Hyperparameter Tuning:*** Extensive computational resources are required to tune hyperparameters effectively.

Future Perspectives and Improvements:

- ***Automated Hyperparameter Optimization:*** Leveraging methods like Bayesian optimization for more efficient lambda selection.
- ***Hybrid Models:*** Combining ridge regression with other machine learning approaches to balance bias and variance more effectively.

F. TEXT INTERVAL TIME-SERIES REGRESSION

Challenges:

- ***Non-Stationarity:*** Text data over time may exhibit trends and seasonality, complicating modeling.
- ***Sparse Temporal Data:*** Text data may not be uniformly sampled over time, leading to gaps and irregular intervals.

Future Perspectives and Improvements:

- ***Advanced Time Series Models:*** Developing or adapting models that can handle non-stationarity and irregular sampling directly.
- ***Integration with NLP Trends:*** Utilizing state-of-the-art NLP models to enhance feature extraction and representation over time.

VII. CONCLUSION

This survey on text regression analysis has introduced a detailed methodological taxonomy that refines the categorization of algorithms into specific techniques and overarching methodology categories. By implementing this granular taxonomy, our study provides a clearer understanding of the nuances among algorithms and offers insights into the appropriate application of various text regression techniques. This clarity is crucial for researchers aiming to select the most suitable algorithms for their specific data and analysis needs.

Our empirical and experimental evaluations reveal significant insights into the performance of text regression methods across different datasets. For instance, while Ridge Regression demonstrated the best overall performance in handling multicollinearity and providing stable predictions, logistic regression outperformed linear regression in datasets with categorical outcomes. This indicates that the choice of regression method should consider the specific characteristics of the dataset to optimize accuracy.

Furthermore, the experimental results underscored the effectiveness of multiple regression methods over simple regression techniques, particularly in complex datasets where multiple predictors are involved. However, challenges such as overdispersion in Poisson Regression and the limitations of Time Series Regression with non-stationary data highlight the importance of understanding each method's inherent limitations and strengths.

In conclusion, our structured approach to classifying text regression algorithms, coupled with comprehensive empirical and experimental insights, equips researchers with a deeper understanding of these methods. This foundation allows for more informed decision-making, ultimately advancing the field of text analysis by improving the accuracy and efficacy of text regression techniques. As text data continues to grow in size and complexity, such nuanced understanding and methodological precision will be paramount in harnessing its full potential.

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